

■ Zerodha Phase 1

External Engines & Systems Report

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Executive Summary

This report documents all external engines, cloud services, and third-party systems integrated into the Zerodha Phase 1 trading project. The system comprises 8 live trading instances, multiple calibration and backtesting engines, and supporting infrastructure.

IMPORTANT: All engines are read-only/paper trading. `LIVE_TRADING_ENABLED` is False everywhere. No order-placement code exists in production paths.

1. Cloud Infrastructure

AWS EC2

Property	Value
Purpose	Distributed calibration for parameter optimization
Instance Type	c6i.4xlarge (16 CPU cores)
Time Savings	Reduces 24h calibration → 45 min
Cost	~\$5 (~\$500) per run
Documentation	AWS_CALIBRATION_GUIDE_STEPBYSTEP_20260830.md

2. Data Storage & Caching

Redis

- Real-time state management and circuit breaker logic
- Trading state (allowed/halted)
- Daily loss tracking
- Maximum drawdown tracking
- Volatility and correlation monitoring
- Stress factor management

3. Broker Integration

Zerodha Kite API

- Indian stock market (NSE - National Stock Exchange)
- Live market data streaming
- Order placement (paper trading mode)
- Position management
- Account info and holdings
- Daily token refresh via `get_kite_access_token.ps1`

4. Data Analytics & ML Libraries

Library	Purpose	Usage
NumPy	Numerical computing	Matrix operations, statistics
Pandas	Data frames	OHLCV manipulation, backtesting
Scikit-learn	Machine learning	Feature scaling, training
XGBoost	Gradient boosting	Parameter optimization
CVXPY	Convex optimization	Portfolio optimization
CVXPortfolio	Portfolio analytics	Cost modeling, risk analysis
TA-Lib	Technical indicators	ATR, RSI, MACD calculations

5. Core Trading Engines

Institutional Engine V3.4 (Variants)

- V3.4 - Baseline production version
- V3.4 B12/B13 - Resilience variants
- V3.4 P01D - Sovereign authorization
- V3.4 P02 - Multi-position candidate

Key Components:

- Trade execution pipeline
- Order status tracking
- Position management
- P&L; calculation
- Risk management

6. Calibration Engines

Parameter Optimization

Parameters Optimized: 33 total

- Entry and exit thresholds
- Position sizing rules
- Risk limits
- ATR scaling factors

Optimization Methods:

- Walk-forward analysis
- Bootstrap sampling
- Cross-sectional optimization
- Risk-adjusted metrics

7. Backtesting Engines

Multi-Period & Historical Testing

Test Periods Supported:

- 1 Month (M)
- 1 Year (Y)
- 2 Years (2Y)
- 3 Years (3Y)

Metrics Calculated:

- P&L; (Realized, Unrealized, MTM)
- Win Rate / Sharpe Ratio
- Drawdown Analysis
- Transaction Costs
- Risk-Adjusted Returns

8. Live Trading Engines (8 Instances)

#	Engine Name	Universe	Mode
1	Chart Studies Monitor	5 symbols	Live
2	P02 Live Scan	Nifty 50	Live
3	Read-Only Shadow	Cross-sectional 12_1	Shadow
4	ORB Shadow Collector	ORB scan	Shadow
5	P01D Entry Gate	V3.4 auth pipeline	Dry-run
6	Dashboard Daemon	3 dashboards	Monitor
7	V11 Bridge - Terminal A	19 symbols	Shadow
8	V11 Bridge - Terminal B	50 symbols	Shadow

9. Monitoring & Logging

Observability

- v34_observatory_v3.py - Real-time observation system v3
- v34_observatory_v4.py - Enhanced observation system v4
- Local HTML dashboards (auto-refresh every 2 min)
- Alert aggregation and audit logging

10. Summary: External Engines by Category

Category	Engine/Service	Key Library/Tool
Cloud Computing	AWS EC2	c6i.4xlarge (16 cores)
Caching/State	Redis	redis-py
Broker	Zerodha Kite API	NSE exchange
Data Analysis	NumPy, Pandas	Numerical computing
ML/Optimization	Scikit-learn, XGBoost, CVXPY	Advanced tuning
Technical Analysis	TA-Lib	ATR, RSI, MACD
Trading Core	Institutional Engine V3.4	Order execution
Backtesting	Custom engines	Multi-period validation

Important Notes & Requirements

- **Paper Trading Only:** All engines run in paper/shadow mode. No live money deployed.
- **Daily Token Refresh:** Zerodha Kite API tokens expire daily. Must run `get_kite_access_token.ps1` every morning.
- **Duplicate Prevention:** If the same engine script runs twice, silent file corruption occurs. Always verify running processes.
- **Separate Windows Required:** Engines 1-5 and 7-8 each need their own PowerShell window with `KITE_ACCESS_TOKEN` set.
- **EOD Handling:** Each engine handles its own end-of-day statement automatically after 15:30 IST.

Research & Strategy Variants

Walk-Forward Strategies:

- Anchored Walk-Forward (V10)
- Momentum Walk-Forward (V11, V12)
- Extended History Validation (V13)

Risk-Adjusted Variants:

- V14 - Realistic Costs
- V15 - Risk Config Realignment
- V16 - Position Count Sweep
- V17 - Multi-Position CNC Engine

Conclusion

The Zerodha Phase 1 project integrates a comprehensive ecosystem of external engines and systems for trading, calibration, and analysis. The architecture leverages cloud computing for fast calibration, Redis for state management, Zerodha's Kite API for market access, and Python's scientific stack for analysis. With 8 live trading instances running in shadow/paper mode and multiple backtesting frameworks, the system provides robust validation and monitoring capabilities for algorithmic trading strategy development.

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